# Finclub Open Project (2025)

## Credit Card Behavior Score Prediction Using Classification and Risk-Based Techniques

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### 1. Problem Statement

The goal of this project is to predict whether a credit card user will default in the following month based on their historical financial behavior. This prediction is framed as a binary classification task (1 = default, 0 = no default), and the dataset provided includes anonymized customer-level attributes.

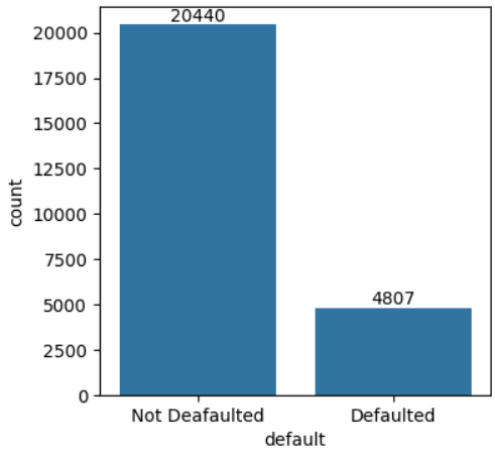
### 2. Dataset Overview

- Train Dataset Size: 25247 rows x 27 columns
- Validation Dataset Size: 5017 rows x 26 columns

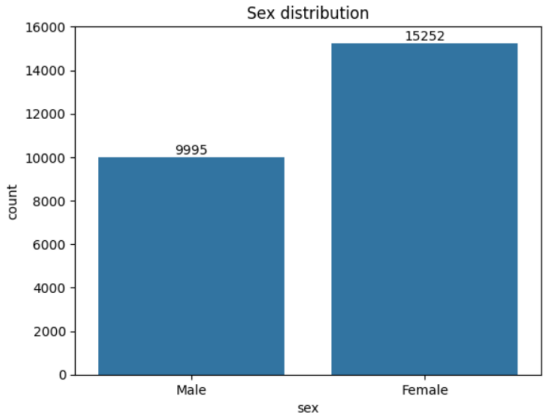
### 3. Exploratory Data Analysis (EDA)

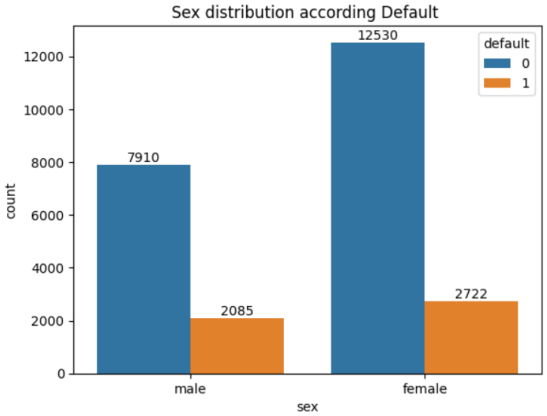
- Checked for null values and handled missing AGE by imputing median
- Visualized target distribution (found slight class imbalance)
- Correlation heatmap showed strong correlation among BILL_AMT and PAY_AMT columns

- **Results from Default Distribution Plot:** This plot shows the distribution of the target variable 'default'. It indicates the number of customers who defaulted in the next month (labeled as 1) and those who did not (labeled as 0). This helps understand the class imbalance in the dataset.

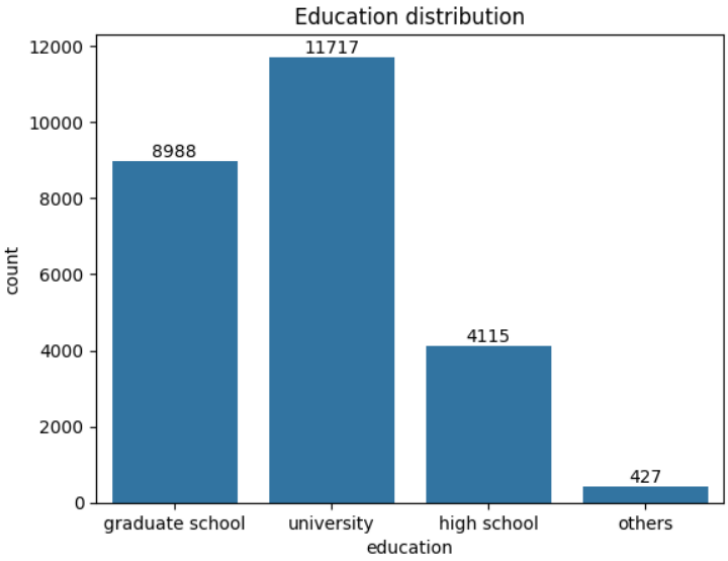


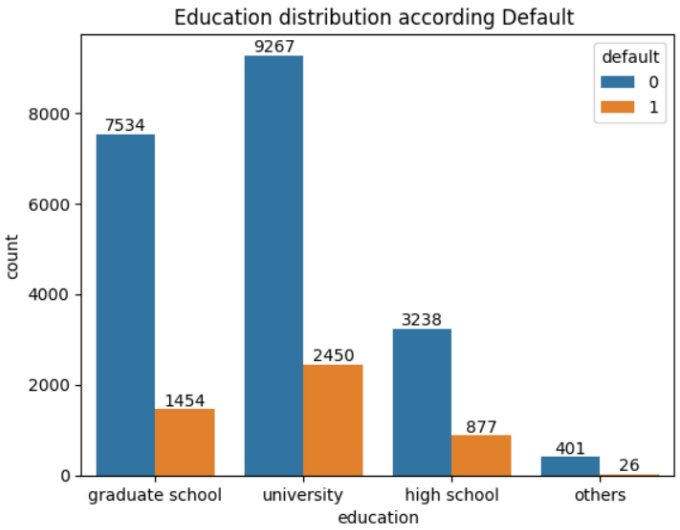

- **Results from Sex Distribution Plots:** The first plot shows the overall distribution of sex in the dataset, indicating the number of male and female customers. The second plot breaks down the sex distribution by default status, allowing us to see if there's a difference in the default rate between male and female customers



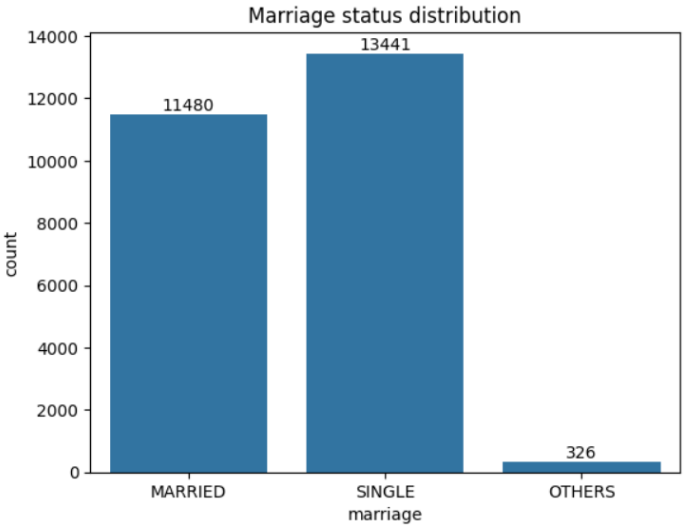



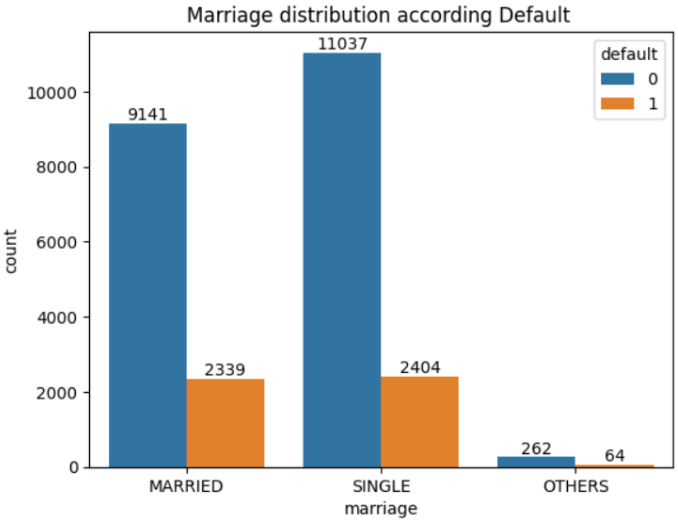
- **Results from Education Distribution Plots:** The first plot shows the overall distribution of education levels among the customers. The second plot shows the education distribution according to default status, which helps in understanding if certain education levels are more prone to defaulting.



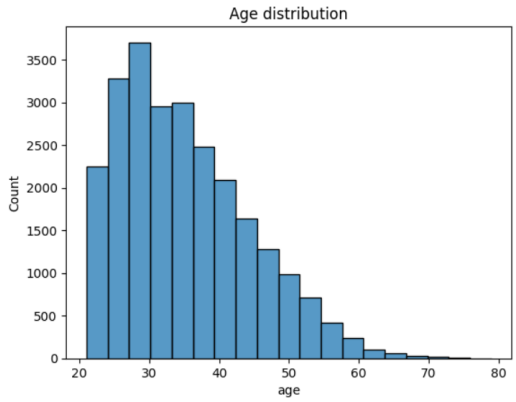



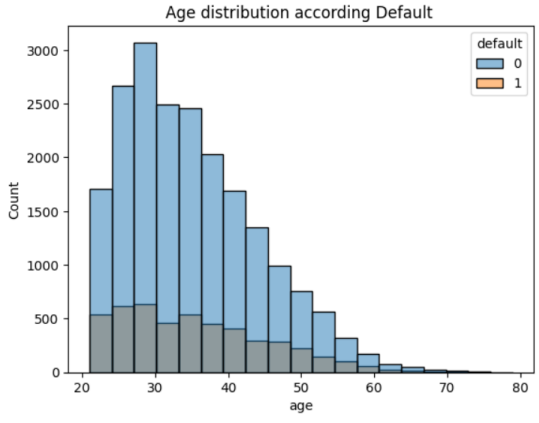

- **Results from Marriage Status Distribution Plots:** The first plot displays the distribution of marriage statuses (MARRIED, SINGLE, OTHERS) in the dataset. The second plot shows the marriage status distribution based on default status, allowing for an analysis of whether marriage status influences the likelihood of default.



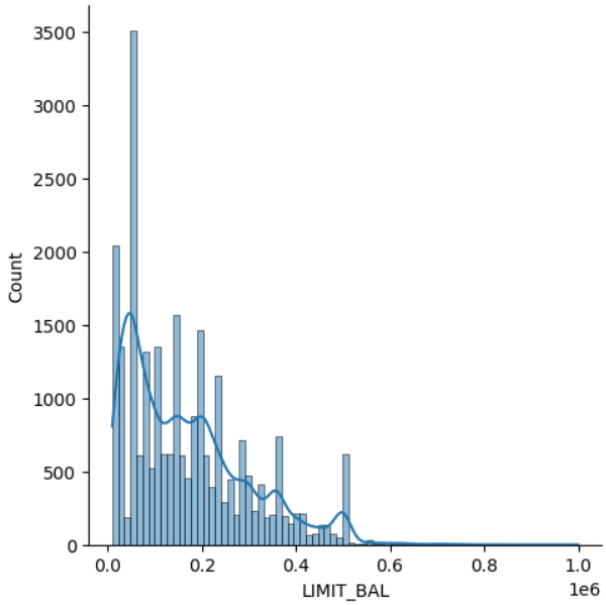




- **Results from Age Distribution Plots:** The first histogram shows the overall distribution of age among the customers. The second histogram shows the age distribution separated by default status, which helps to identify if certain age groups have a higher or lower default rate.

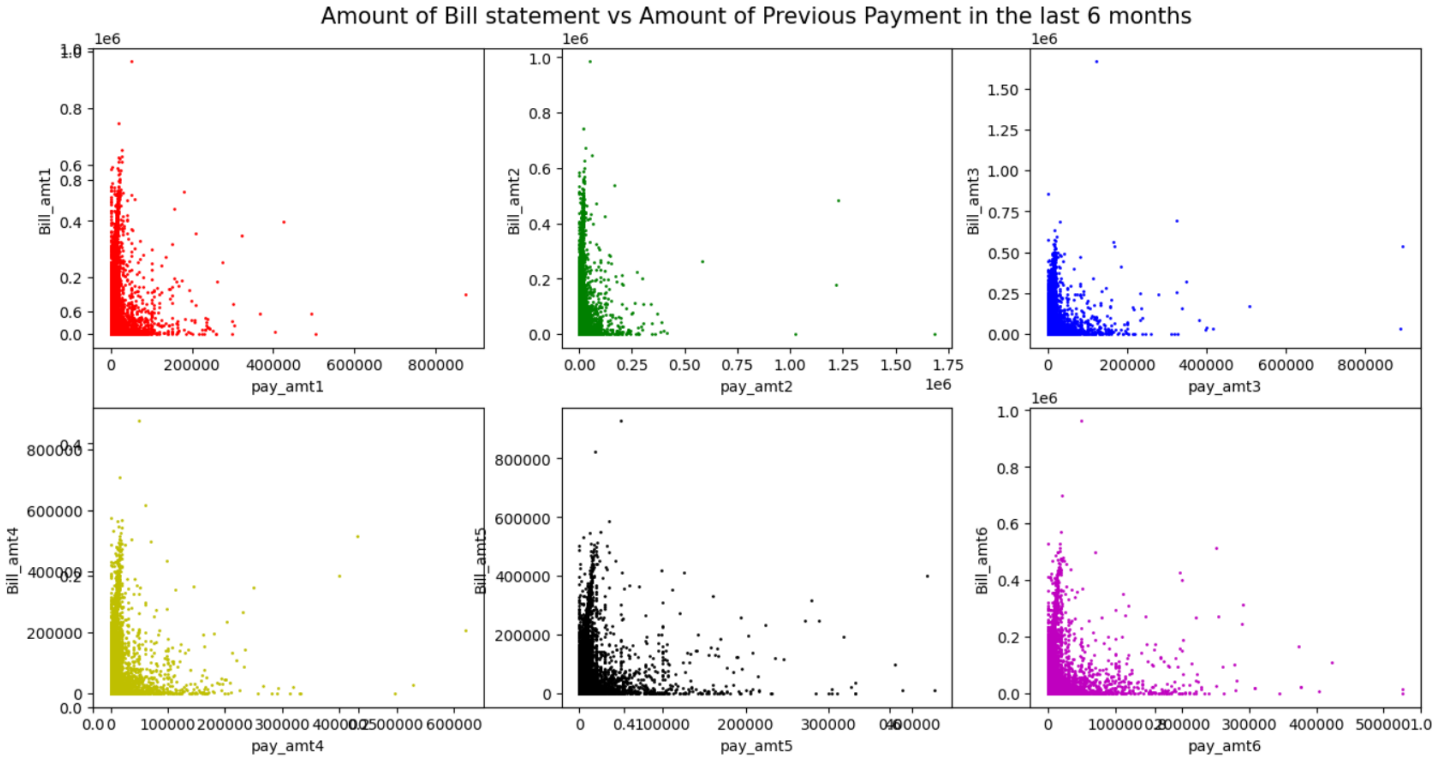


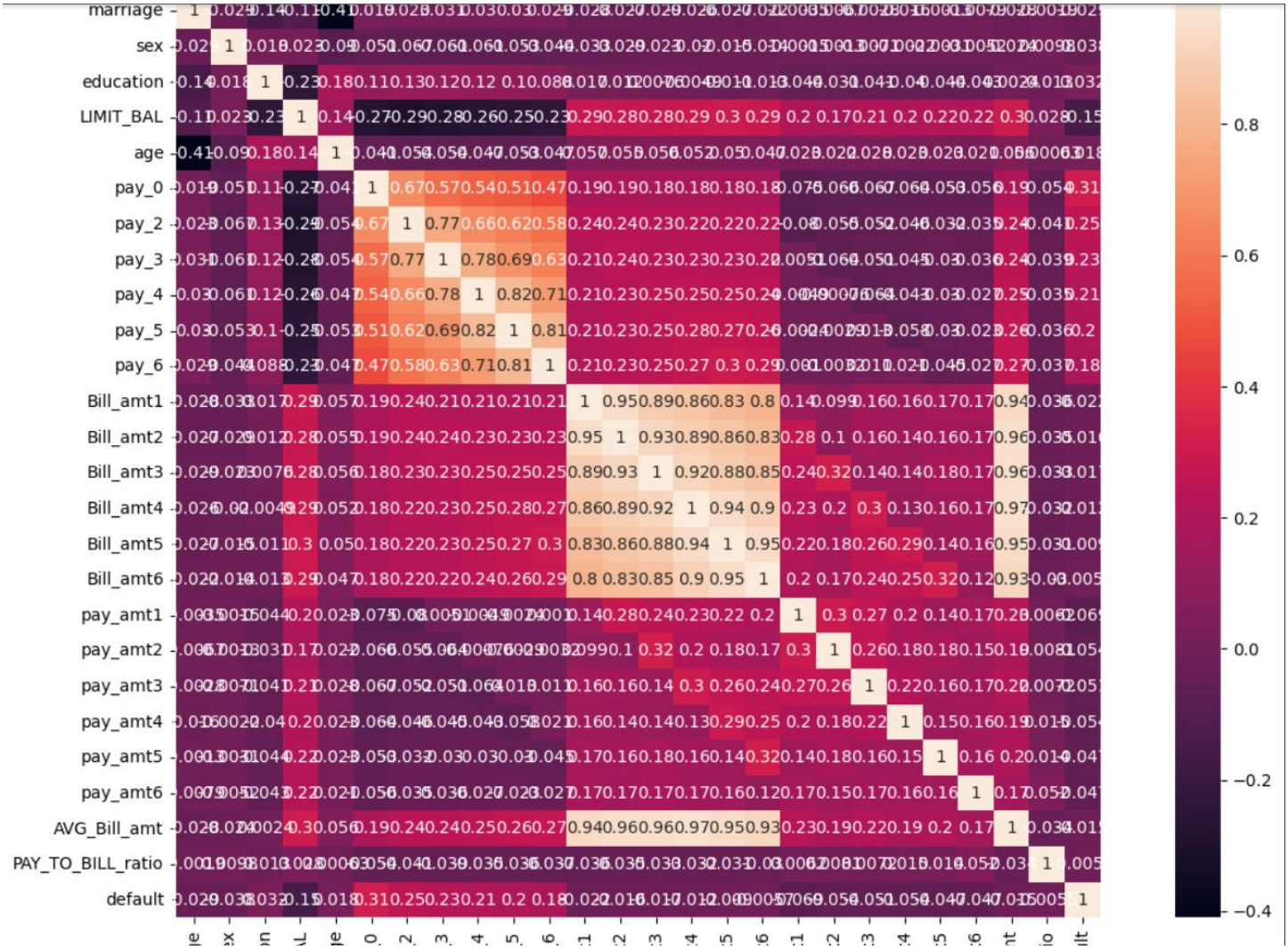
- **Results from LIMIT_BAL Distribution Plot:** This distribution plot shows the distribution of the credit limit (LIMIT_BAL) among the customers. It helps to understand the range and frequency of different credit limits in the dataset.




- **Results from Bill Amount vs. Payment Amount Scatter Plots:** These scatter plots visualize the relationship between the bill statement amount and the previous payment amount for each of the past six months. Each subplot represents a different month. These plots can reveal patterns such as whether higher bill amounts are associated with higher payment amounts, or if there are clusters of customers with high bills and low payments, which might indicate a higher risk of default.

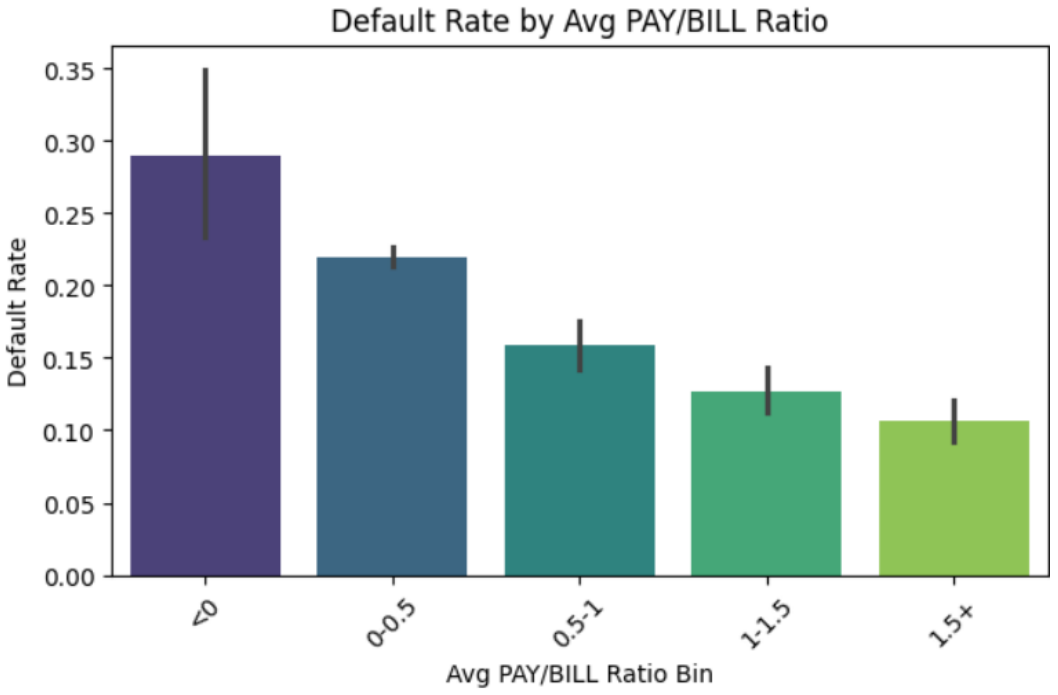


**Correlation Map**

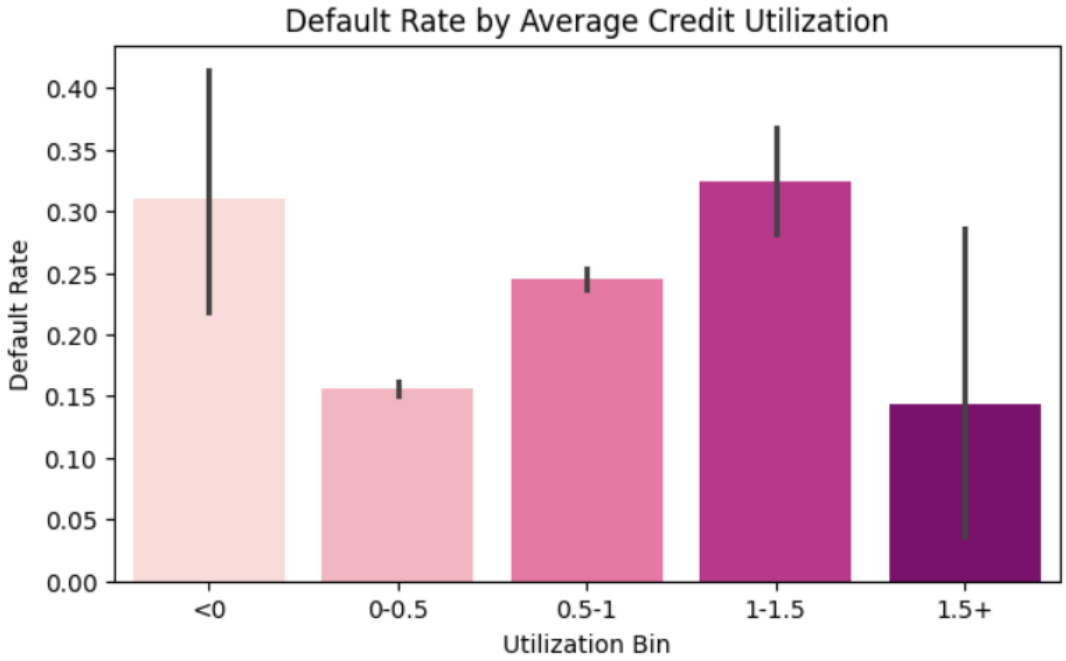



## 4. Financial insights

- **Payment-to-Bill Ratio Trend Analysis:** This bar plot shows the default rate for different bins of the average payment-to-bill ratio. It helps to understand if customers with a lower average payment-to-bill ratio (meaning they pay back a smaller proportion of their bill amount on average) have a higher default rate.



- **Applying the concept of credit utilization:** This bar plot shows the default rate for different bins of average credit utilization. It helps to understand if customers with higher average credit utilization (meaning they are using a larger proportion of their available credit) have a higher default rate. The plot visually indicates how default risk changes with different levels of credit utilization




## 5. Data Preprocessing

- Missing Values: Imputed missing AGE values with median
- Label Encoding: Encoded categorical variables (SEX, EDUCATION, MARRIAGE)
- Scaling: Used StandardScaler
- Train-Test Split: 80-20 split

## 6. Model Development & Evaluation

Models Tried:
- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- XGBoost with Hyperparameter Tuning

Evaluation Metric:
F2 Score was selected as the main evaluation metric because recall is more important than precision in credit default prediction (missing defaulters is costlier than false alarms).

Performance Summary:

| Model | Accuracy | F2 Score |
|---|---|---|
| Logistic Regression | 41.88% | 0.5599 |
| Decision Tree | 57.8% | 0.5707 |
| Random Forest | 56.04% | 0.6023 |
| XGBoost | 51.9% | 0.5858 |
| XGBoost (Tuned) | 56.12% | 0.5959 |

**Reasoning for Selecting the Best Model:**

When dealing with imbalanced datasets, like this credit default prediction problem where the number of non-defaulters is significantly higher than defaulters, Accuracy can be a misleading metric. A model could achieve high accuracy by simply predicting the majority class (non-defaulters) for most cases.

The F2 score is a better metric in this scenario because it weights Recall twice as much as Precision. Recall (also known as sensitivity) measures the model's ability to correctly identify all positive instances (defaulters). In the context of credit default prediction, correctly identifying potential defaulters is often more critical than being perfectly precise (avoiding false positives), as missing a defaulter can lead to significant financial loss.

Based on the F2 scores from the table above, the Random Forest model (with a threshold of 0.22, resulting in an F2 score of 0.6023) achieved the highest F2 score among all the trained models. While the Hyperparameter-Tuned XGBoost model also performed well with an F2 score of 0.5959, the Random Forest model showed slightly better performance based on this metric.
**Therefore, the Random Forest model is selected as the best model for this task based on its superior F2 score, indicating a better balance between identifying actual defaulters (Recall) and minimizing false positives (Precision), with a stronger emphasis on Recall.**

## 7. Validation Prediction

- Used the final tuned Random Forest model with a threshold of 0.22 to predict on the validation dataset
- Saved output as predictions_default.csv
- Applied the same preprocessing steps as on training data

## 8. Challenges Faced

- Imputing missing values without data leakage
- Maintaining consistent preprocessing pipeline between train and validation datasets
- Hyperparameter tuning took time due to computational cost